

agar_fresh

- **3 × 250 mL** bottles, plus spare empties
- **1 level tablespoon** powder per bottle
- DI to the neck line · cap · shake
- Cooker: **Beef** (small) / **Sterilize** (large)
- **20 min** high pressure · natural release
- Cool to **~55 °C**

Preparation of 2YT Broth (250 mL bottles)

broth_2yt

- **4 × 250 mL** bottles
- Premixed 2YT powder — **check the scoop on the tub**
- or components: **16 / 10 / 5 g/L** tryptone / yeast ext / NaCl
- DI to the neck line · shake until clear
- **20 min** high pressure · natural release
- Cooker holds **5 bottles** → 3 agar + 4 broth = **2 runs**

abx_carb

Carbenicillin

- **100 mg/mL**
- $\mu\text{L} = \text{mg} \times 10$
- Solvent: **water**
- Scoop **~50 mg** · tare first · record the mass

abx_spec

Spectinomycin

- **100 mg/mL**
- $\mu\text{L} = \text{mg} \times 10$
- Solvent: **water**
- Scoop **~50 mg** · tare first · record the mass

abx_kan

Kanamycin

- 25 mg/mL
- $\mu\text{L} = \text{mg} \times 40$
- Solvent: water
- Scoop ~25 mg, not 50 — 50 will not fit the tube
- Protect from light

abx_erm

Erythromycin

- **100 mg/mL**
- $\mu\text{L} = \text{mg} \times 10$
- Solvent: **ethanol** — will not go into water at all
- Scoop **~50 mg** · vortex until clear, it is slow
- Crashes out of hot agar, then redissolves

agar_remel

- **1 × 250 mL** bottle
- Loosen the lid **a quarter turn**
- **30–60 s bursts** · rest between
- Stop at boiling
- Cool to **~55 °C**

Pouring many Carb petri dishes

pour_many_plates

Carbenicillin · 1 stack

- **~25 mL** per plate · **~10 plates** per 250 mL
- 250 mL agar + **250 µL** carb 1000×
- Add at **~55 °C** · swirl, do not shake
- Chemical hood · burner lit
- **Red stripes** · “Carb” on the bottom

Pouring a few Erm petri dishes with conical

pour_few_plates1

Erythromycin · 3 plates

- $3 \times 25 \text{ mL} = \mathbf{75 \text{ mL}} + \mathbf{75 \text{ }\mu\text{L}}$ erm 1000×
- **75 mL will not fit one 50 mL conical** — two, or pour thinner
- Fresh conical, this combination only
- Add at **~55 °C** · swirl, do not shake
- No stripe convention exists — **pick one**

Pouring a few spec petri dishes with bottle

pour_few_plates2

Spectinomycin · 4 plates

- $4 \times 25 \text{ mL} = \mathbf{100 \text{ mL}} + \mathbf{100 \text{ }\mu\text{L}}$ spec 1000×
- Sterile **100 mL bottle**, this combination only
- Add at **~55 °C** · swirl, do not shake
- **Black stripes** · “Spec” on the bottom

Pouring and Spreading Large Petri Dishes (150 mm)

pour_large_plate

150 mm plates

- **~50 mL** per plate · **5 plates** per 250 mL bottle
- Count: **---**
- 1000× rule: **X μ L per X mL**
- Add at **~55 °C** · swirl, do not shake

Pouring a few kan petri dishes

pour_kan

NOT FILMED · Kanamycin · 3 plates

- $3 \times 25 \text{ mL} = 75 \text{ mL} + 75 \mu\text{L kan } 1000\times$
- Add at **$\sim 55^\circ\text{C}$** · swirl, do not shake
- **Green stripes** · “Kan” on the bottom

Pouring a few plain petri dishes

pour_plain

NOT FILMED · no antibiotic · 3 plates

- $3 \times 25 \text{ mL} = \mathbf{75 \text{ mL}}$, nothing added
- The no-drug control — proves the others select
- No stripes

Streaking control strain

ctrl_streak

4 control strains · from the -80 box

- Plates: **carb** · **spec** · **kan** · **erm**
- 3 plates, all 4 strains
- Each strain on its own plate; **carb strain onto spec as the neg**
- Streak **for isolation** · scrape the stock, do not thaw it
- Invert · **37 °C** overnight

Label plates, thaw cells, incubate dna

ctrl_transform1

4 reactions · carb · kan · spec · erm

- Label the plate bottoms first
- EchoTherm **A = 4 °C**, **B = 42 °C**
- Cells on block A · thaw ~**30 s** · add **25 µL KCM**
- **0.5 µL** miniprep into **10 µL** cells
- **10 min at 4 °C** ← stop here
- Same for all four — the fork is at step 2

Heat shock and plate

ctrl_transform2a

NO RESCUE · carb only · 1 reaction

- **90 s at 42 °C → 1 min at 4 °C**
- Carbenicillin → **plate straight away**
- Glass beads · shake side to side, not around
- Invert · **37 °C** overnight

Heat shock and rescue

ctrl_transform2b

RESCUE · kan · spec · erm · 3 reactions

- **90 s at 42 °C → 1 min at 4 °C**
- **Add 200 µL 2YT · move to a 1.5 mL tube**
- **Shake 37 °C for 1 h**
- **Do not plate yet → ctrl_transform3a**

Plate from a rescue

ctrl_transform3a

kan · spec · erm · 3 reactions

- Plate **all** the liquid
- Glass beads · shake side to side, not around
- **Dry uncovered** at the sterile bench until not glossy
- — 200 μ L will bleed otherwise
- Invert · **37 °C** overnight

Label plates, thaw cells, incubate dna

bestP_transform1

7 reactions • all on carb

- **One 100 μ L aliquot** for all seven
- **16 μ L cells + KCM + 0.5 μ L DNA** each
- pJ01 / pJ12 / pJ19 $\rightarrow$ **Training Box**
- 4 pP6 clones $\rightarrow$ **TPcon6B box**
- **10 min at 4 $^{\circ}$ C** $\leftarrow$ stop here

Heat shock and plate

bestP_transform2a

NO RESCUE · 7 reactions · all on carb

- **90 s at 42 °C → 1 min at 4 °C**
- Carbenicillin → **NO rescue**
- Glass beads · shake side to side, not around
- Invert · **37 °C** overnight